

PCCS4205	Database Management System	3-0-0	Credit-3
Module I: (10 Hours) Introduction to database Systems, advantages of database system over traditional file system, Basic concepts & Definitions, Database users, Database Language, Database System Architecture, Schemas, Sub Schemas, & Instances, database constraints, 3-level database architecture, Data Abstraction, Data Independence, Mappings, Structure, Components & functions of DBMS, Data models. Module II: (10 Hours) Entity relationship model, Components of ER model, Mapping E-R model to Relational Schema. Storage Strategies: Detailed Storage Architecture, Storing Data, Magnetic Disk, RAID, Other Disks, Magnetic Tape, Storage Access, File & Record Organization, File Organizations & Indexes, Order Indices, B+ Tree Index Files, Hashing, Data Dictionary. Module III: (10 Hours) Relational Algebra (RA), Tuple Relational Calculus (TRC) and Domain Relational Calculus (DRC), Relational Query Languages: SQL and QBE. Database Design: Database development life cycle (DDLC),			

Automated design tools, Functional dependency and Decomposition, Join strategies, Dependency Preservation & lossless Design, Normalization, Normal forms: 1NF, 2NF, 3NF, and BCNF, Multi-valued Dependencies, 4NF & 5NF. Query processing and optimization: Evaluation of Relational Algebra Expressions, Query optimization, Query cost estimation.

Module IV: (10 Hours)
Transaction processing and concurrency control: Transaction concepts, properties of transaction, concurrency control, locking and Timestamp methods for concurrency control schemes. Database Recovery System: Types of Data Base failure & Types of Database Recovery, Recovery techniques. Fundamental concepts on Object-Oriented Database, Object relational database, distributed database, Parallel Database, Data warehousing & Data Mining, Big Data and NoSQL.

Text Books:

1. Database System Concepts by Sudarshan and Korth, 6th edition, McGraw-Hill Education.
2. Fundamentals of Database System by Elmasari & Navathe, Pearson Education.
3. Fundamentals of Database Management System by Gillenson, Wiley India.

References Books:

1. Database Management Systems by Ramakrishnan, McGraw-Hill Education.
2. Database management system by leon & leon, Vikas publishing House.
3. Database System: Concept, Design & Application by S.K. Singh, Pearson Education.
4. An introduction to Database System – Bipin Desai, Galgotia Publication

Registration no:

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B.Tech. CSE

Subject Code: PCCS4205

45-52

4th Semester Regular/ Back Paper Examination 2021-22**Subject Name: Database Management System****Branch: Computer Science & Engg.****Time: 3 Hours****Max Marks: 60****Question Code:****Answer Question No.1 which is compulsory and any five from the rest.****The figures in the right-hand margin indicate marks.**

- Q1/** **Answer the following questions:** **(1x10)**
- ✓ Explain 3-tier architecture of DBMS.
 - ✓ Are NULL values in a database same as blank space or zero.
 - ✓ Mention the issues with traditional file-base system that make DBMS a better choice.
 - ✓ Explain different types of relationships amongst tables in a DBMS.
 - ✓ Differentiate between intention and extension in a database.
 - ✓ Describe the different levels of data abstraction in a data base.
 - ✓ Explain Domain and Tuple Relational calculus with examples.
 - ✓ What do you understand by aggregation and atomicity?
 - ✓ Differentiate between Functional Dependency and Multivalued Dependency.
 - ✓ What are the different integrity rules present in the DBMS?
- Q2**
- a) What is a participation role? When is it necessary to use role names in the description of relationship types? **(5)**
 - b) What is meant by a recursive relationship type? Give some examples of recursive relationship types. **(5)**
- Q3/**
- ✓ a) What is the difference between a specialization hierarchy and a specialization lattice? **(5)**
 - ✓ b) Design an EER schema for a database application that you are interested in. Specify all constraints that should hold on the database. Make sure that the schema has at least five entity types, four relationship types, a weak entity type, a superclass/subclass relationship, a category, and an n -ary ($n > 2$) relationship type. **(5)**
- Q4/**
- ✓ a) Consider the following relations for a database that keeps track of business trips of salespersons in a sales office:
 SALESPERSON (Ssn, Name, Start_year, Dept_no)
 TRIP(Ssn, From_city, To_city, Departure_date, Return_date, Trip_id)
 EXPENSE(Trip_id, Account#, Amount)
 A trip can be charged to one or more accounts. Specify the foreign keys, Candidate keys and Primary keys for this schema, stating any assumptions you make. **(5)**
 - ✓ b) What is union compatibility? Why do the UNION, INTERSECTION, and DIFFERENCE operations require that the relations on which they are applied be union compatible? **(5)**

Q5 a) Consider the following MAILORDER relational schema describing the data for a mail order company. (5)

PARTS(Pno, Pname, Qoh, Price, Olevel)

CUSTOMERS(Cno, Cname, Street, Zip, Phone)

EMPLOYEES(Eno, Ename, Zip, Hdate)

ZIP_CODES(Zip, City)

ORDERS(Ono, Cno, Eno, Received, Shipped)

ODETAILS(Ono, Pno, Qty)

Qoh stands for *quantity on hand*: the other attribute names are self-explanatory.

Specify and execute the following queries using the RA expression on the MAILORDER database schema.

- Retrieve the names of parts that cost less than \$20.00.
- Retrieve the names and cities of employees who have taken orders for parts costing more than \$50.00.
- Retrieve the pairs of customer number values of customers who live in the same ZIP Code.

b) What are the binary operators in relational algebra? (5)

- Q6 a) (a) Discuss the correspondences between the ER model constructs and the relational model constructs. Show how each ER model construct can be mapped to the relational model and discuss any alternative mappings. (5)
- b) (b) Discuss the options for mapping EER model constructs to relations, and the conditions under which each option could be used. (5)

- Q7 a) Define *Boyce-Codd normal form*. How does it differ from 3NF? Why is it considered a stronger form of 3NF? (5)
- b) Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies $F = \{ \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$. What is the key for R ? Decompose R into 2NF and then 3NF relations. (5)

Q8 Write short notes on any two: (5x2)

- Anomalies In DBMS.
- Physical Data Independence and Logical Data Independence.
- Transitive Dependency
- De Normalization



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B.Tech.

Subject Code: PCCS4205

4th Semester Regular/ Back Paper Examination 2022-23

Subject Name: Database Management System

Branch: Computer Science & Engineering

Time: 3 Hours

Max Marks: 60

Question Code:

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right-hand margin indicate marks.

- Q1 Answer the following questions: (1x10)
- What does defining, manipulating and sharing of a database mean?
 - What is the difference between a database schema and a database state?
 - Justify why handling of null values is difficult?
 - Explain entity integrity and referential integrity constraints.
 - How are the OUTER JOIN operations different from INNER JOIN operations?
 - Explain the differences among a relationship instance, a relationship type and a relationship set.
 - What do you understand by attribute closure?
 - List the informal guidelines for a relational schema design.
 - When does a dangling tuple occur? Give an example.
 - How is the concept of serializability useful for concurrency control?
- Q2 a) What are database utilities? What are their functions? (5)
- (b) What is the difference between the two-tier and three-tier client/server architectures? (5)
- Q3 a) Specify the following relational algebra operations in both tuple and domain relational calculus (5)
- $R(A, B, C) * S(C, D, E)$
- $R(A, B, C) = S(A, B, C)$
- b) When is a query language called relationally complete? (5)
- Q4 a) Discuss user-defined and predicate-defined subclasses and identify the differences between the two. (5)
- b) Prove that any relation schema with two attributes is in BCNF. (5)
- Q5 Consider the following relation for published books: (10)
- BOOK (Book-title, Author-name, Book-type, List-price, Author-affil, Publisher)
- Suppose the following dependencies exist :
- Book-title $\rightarrow$ Publisher, Book-type
- Book-type $\rightarrow$ List-price
- Author-name $\rightarrow$ Author-affil
- What normal form is the relation in?
- Apply normalization until you cannot decompose the relations further. State reasons behind each decomposition.



- Q6** a) What is the lossless join property of a decomposition? Why is it important? (5)
- (b) Define join dependencies and fifth normal form. Why is 5 NF also called project-join normal form (PJNF) (5)
- Q7** a) List the search algorithms that can be used to implement a SELECT Operation. Mention the conditions under which these algorithms can be used. (5)
- b) What is schedule? Define the concepts of recoverable, cascadeless and strict schedules and compare them in terms of their recoverability. (5)
- Q8** Write short notes on any two: (5x2)
- a) Protocols for deadlock prevention
 - b) System Lock Tables
 - c) Query graph
 - d) Dirty Read problem



INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG
4TH Semester Regular/ Back Paper Examination AY 2023-24

Total no. of Pages:
02

Branch	CSE	Program	B. Tech		
Subject Name	DATABASE MANAGEMENT SYSTEM	Semester	4 TH		
Subject Code	PCCS4205	Month/Year	2023-24		
Time: 03 Hrs	Answer Question number 1 which is compulsory and any five from the rest.	Maximum Marks	60		
Each Question Carry Marks (M) as Indicated in the right-hand Margin					
Q's	Questions	M	BL	CO	PO
1. a	Difference between a database schema and a database state.	01	L1	1	1
b	Explain why handling null values is difficult.	01	L2	1	1
c	Discuss various types of inner join operations.	01	L1	2	2
d	Compare the difference between stored and derived attributes.	01	L3	3	3
e	Explain the informal guidelines for a relational schema design.	01	L6	3	3
f	What is the lossless join property of a decomposition? Why is it important?	01	L6	3	3
g	List the search algorithms that can be used to implement a SELECT operation.	01	L2	3	3
h	What are the conditions that lead to the two schedules being view equivalent.	01	L6	4	4
i	Discuss the actions taken by the read-item and write-item operations on a database.	01	L5	4	4
j	Explain the transparency features of a Distributed Database Management System.	01	L4	5	5
2. a	What are the characteristics of relations that make it different from ordinary tables and files.	05	L3	1	1
b	How does tuple relational calculus differ from domain relational calculus?	05	L2	1	1
3. a	Consider the following relational database schema consisting of the four relation schemas: Passenger (Pid, Pname, Pgender, Pcity) Agency (Aid, Aname, Acity) Flight (Fid, Fdate, time, src, dest) Booking (Pid, Aid, Fid, Fdate) Answer the following questions using relational algebra queries. (a) Get the complete details of all flights to New Delhi. (b) Get the details about all flights from Chennai to New Delhi. (c) Find only the flight numbers for passenger with Pid 124 for flights to Chennai before 06:11:20 (d) Find the agency names for agencies who do not	05	L5	2	2

	have any bookings for passenger with id 123.				
b	Write the above queries using Tuple Relational Calculus.	05	L2	2	2
4.a	Given the following requirements for a simple database for the National Hockey League (NHL). NHL has many teams, each team has a name, a city, a coach, a captain and a set of players, each player belong to only one team, each player has a name, a position (left wing or goalie), a skill level and a set of injury records, a team captain is also a player, a game is played between two teams (host-team and guest-team) has a date and a score (4 to 2). Construct a clean and concise ER diagram for NHL database, indicate the cardinality mapping and participation details in your diagram.	05	L5	3	3
b	What is the difference between specialization hierarchy and specialization lattice? Explain with an example.	05	L4	3	3
5	You are given the following set of functional dependencies for a relation R (A, B, C, D, E, F), F = {AB → C, DC → AE, E → F} What are the keys of this relation. Is the relation in BCNF, Is the decomposition (A, B, C, D) and (B, C, D, E, F) dependency preserving. Find the 3NF form of the above relation.	05	L3	3	3
b	An organization needs to maintain database having five attributes A, B, C, D, E. These attributes functionally dependent on each other. Let F: (A → BC, D → E, BC → D, A → D) of R (A, B, C, D, E). Find the minima cover F ¹ of F.	05	L3	3	3
6.a	What is query execution plan? Optimize the following query. σ _{salary > 4000} (EMPLOYEE IXI _{ono-Dnumber} DEPARTMENT)	05	L5	4	4
b	How one buffering and caching techniques used by the recovery subsystem. Describe write-ahead logging protocol.	05	L2	4	4
a	Check whether the given schedule S is conflict serializable or not. Draw the precedence graph. S: R ₁ (A), R ₂ (A), R ₁ (B), R ₂ (B), R ₃ (B), W ₁ (A), W ₂ (B).	05	L5	5	5
b	Discuss the timestamp ordering protocol for concurrency control, how does strict timestamp ordering differ from basic timestamp ordering?	05	L2	5	5
Short notes (Any Two)					
8.a	Data Fragmentation.	05	L2	5	5
b	Thoma's write rule.	05	L1	4	4
c	Data marts.	05	L3	3	3
d	Constraints on specialization and generalization.	05	L4	2	2



INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG
4th Semester Regular/ Back Paper Examination AY 2024-25

Total no. of Pages:
02

Branch	COMPUTER SCIENCE & ENGINEERING	Program	B. Tech
Subject Name	DATABASE MANAGEMENT SYSTEM	Semester	4 TH
Subject Code	PCCS4205	Month/Year	MAY/2025
Time: 03 Hrs	Answer Question number 1 which is compulsory and any five from the rest.	Maximum Marks	60

Each Question Carry Marks (M) as Indicated in the right-hand Margin

Q's	Questions	M	BL	CO	PO
1. a	Discuss the capabilities that should be provided by a DBMS.	01	L2		
b	What are database utilities? What are their functions?	01	L1		
c	Explain the entity integrity and referential integrity constraints. Why is each considered important.	01	L2		
d	When a query language is called relationally complete?	01	L2		
e	What is meant by a recursive relationship type? Give some examples of recursive relationship types.	01	L2		
f	Define join dependency and fifth normal form.	01	L2		
g	When does a dangling tuple occur? Give an example.	01	L2		
h	What are the different methods used for implementing joins.	01	L1		
i	What is difference between the constrained write and unconstrained write assumptions? Which is more realistic?	01	L2		
✓ j	Describe shadow paging recovery technique. Under what circumstances does it not require a log?	01	L2		
2. a	Draw an ER diagram for the following problem. A team has a number of players, not all of whom participate in each game. It is desired to keep track of the players participating in each game for each team, the positions they played in that game and the result of the game.	05	L3		
✓ b	What is the difference between specialization hierarchy and a specialization lattice? Identify user defined and attribute defined specialization.	05	L2		
3. a	Write the following queries in relational algebra based on the database schema. Classes (Class, Type, Country, Numgun, Bone, Displacement) Ships (Name, Class, Launched) Battles (Name, Date) Outcomes (Ship, Battle, Result) a) Find the class name and country for all classes with at	05	L3		

	least 10 guns. b) Find the names of all ships launched prior to 1918. c) Find the names of ships sunk in battle and the name of the battle in which they were sunk. d) Find all ships that are used by country India. e) Find names of all ships that begin with letter 'R'.				
b	Write the above (Q3(a)) queries using tuple relational calculus.	05	L5		
4.a	Consider the relation R(A, B, C, D, E) with the following dependencies. $G = \{AB \rightarrow C, CD \rightarrow E, DE \rightarrow B\}$ Find the key for the relation R with FD G normalize the above relation up to BCNF. <i>ABD → C, DLE → E, ADE → B, C</i>	05	L5		
b	Consider the following relation R (Doctor#, patient#, Date, Diagnosis, Treat-code, Charge) Assume that diagnosis is determined (uniquely) for each patient by a doctor. Assume that each treatment code has a fixed charge. Is this relation in 2NF? Normalize to 3NF?	05	L5		
5. a	Consider schedules S3, S4 and S5 below. Determine whether each schedule is strict, cascade less, recoverable or nonrecoverable. S3: $r_1(x); r_2(z); r_1(z); r_3(x); r_3(y); w_1(x); c_1; w_3(y); c_3; r_2(y); w_2(z); w_2(y); c_2;$ S4: $r_1(x); r_2(z); r_1(z); r_3(x); r_3(y); w_1(x); w_3(y); r_2(y); w_2(z); w_2(y); c_1;$ S5: $r_1(x); r_2(z); r_3(x); r_1(z); r_2(y); r_3(y); w_1(x); c_1; w_2(z); w_3(y); w_2(y); c_3; c_2;$	05	L5		
b	What is a serial schedule? What is a serializable schedule? Why is serial schedule considered correct? How serializability useful for concurrency control.	05	L2		
6.a	How does the granularity of data items affect the performance of concurrency control? What factors affect selection of granularity size for data items?	05	L4		
b	What is meant by transaction rollback? What is meant by cascading rollback? Why do practical recovery methods use protocols that do not permit cascading rollback? Which recovery techniques do not require any rollback?	05	L2		
7.a	Mention and briefly explain the control measures that are used to provide security of data in database.	05	L2		
b	What are the goals of data mining?	05	L2		
	Short Notes (Answer Any Two)				
8.a	Characteristics of data warehousing	05	L2		
b	Multiversion concurrency control technique	05	L2		
c	Constraints of specialization and Generalization hierarchies	05	L2		
d	Selectivity and cost estimates in query optimization	05	L2		